BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 041 · Programming Techniques · Scope and Lifetime

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does return do in a method?

Q2. What is a condition-controlled loop?

Q3. Why might binary search outperform linear search?

Q4. Simplify ¬¬B.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — scope

Recall the term — local

Recall the term — class field

Recall the term — shadowing

Recall the term — lifetime

New learning

Explain the important decision — int score = 10; inside a method, int score = 3; refers to different variables in different scopes.

Worked example

Spot two variables with one name

1. Identify the task — Spot two variables with one name

2. Follow the logic — int score = 10; inside a method, int score = 3; refers to different variables in different scopes.

3. Check the outcome — Explain how each step contributes to the result. Test an edge case.

We Do · Practise with guidance

Problem 1

A parameter n is set to 4. What is visible inside the function?

Discuss, then respond

Check your thinking

The parameter n with value 4.

Problem 2

Why avoid excessive global state?

Changes from one part of the program can affect another.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Explain one benefit of local variables. [2]

Question 2

Apply independently
A method declares int total=0. Can main read total directly? [1]

Question 3

Apply independently
Define variable shadowing. [2]

Exit Ticket · Answer before checking

True or false: every local variable is visible throughout the program.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sends a value back to its caller

Q2. Repeats while a test holds

Q3. It halves the remaining range

Q4. B

You Do pages · Independent questions

Q1. They limit scope and reduce unintended interference.

Q2. No.

Q3. A declaration in an inner scope uses the same name as one in an outer scope, hiding it there.

Exit Ticket · Model answer

False. A local variable is visible only within its defined scope.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: